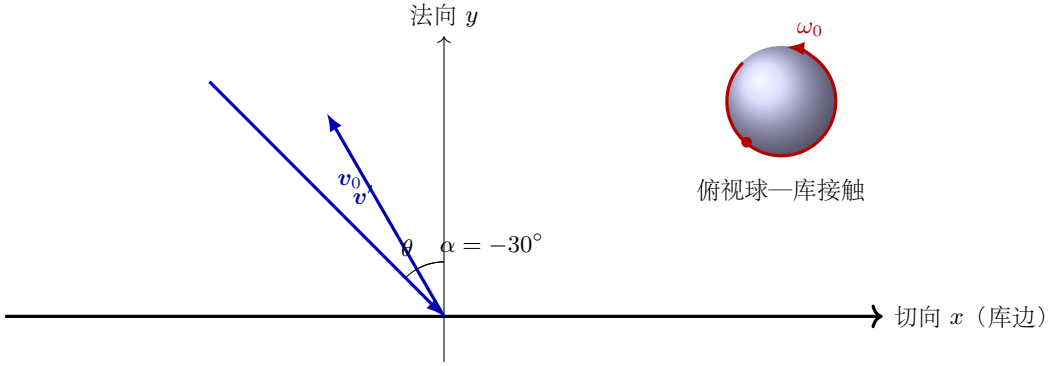
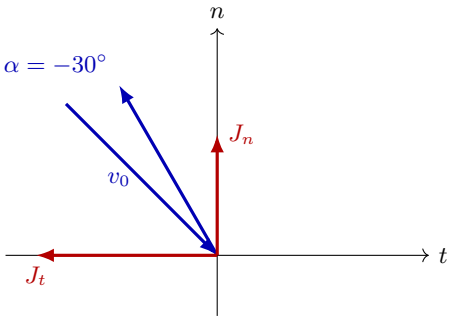
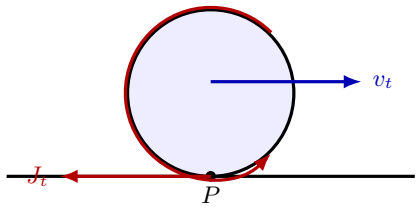

法向 y
ω_0
俯视球—库接触
$\boldsymbol{v}_0$
$\boldsymbol{v}$
θ
$\alpha = -30^\circ$
切向 x（库边）



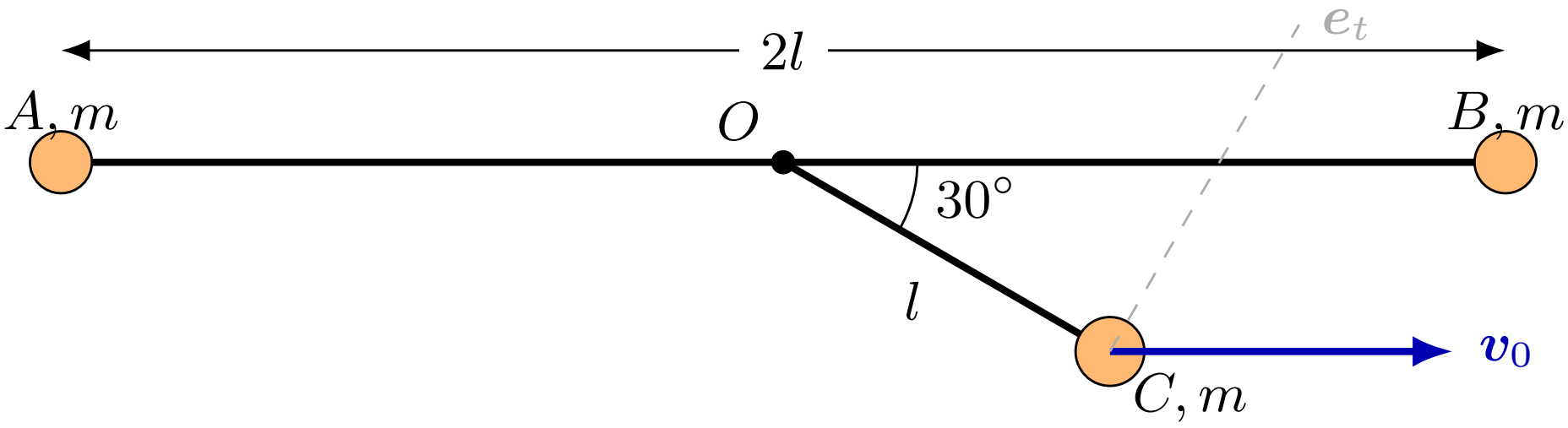
$v_t' = v_t - J_t/m < 0$

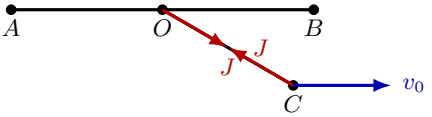
$\omega > 0$：逆时针



$u_P = v_t + R\omega$

$\Delta K = -J_t u_P + \frac{1}{2} J_t^2 (1/m + R^2/I)$


$2l$
e_t
A, m
O
B, m
$30°$
l
$\boldsymbol{v}_0$
C, m



绷直瞬间：$J = mv_0/\sqrt{3}$

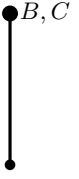
相邻两幅相隔 $4\pi l/v_0$

中间各发生一次 CA 碰撞

四组后复原：$T = 16\pi l/v_0$



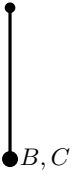
$\theta = 0$



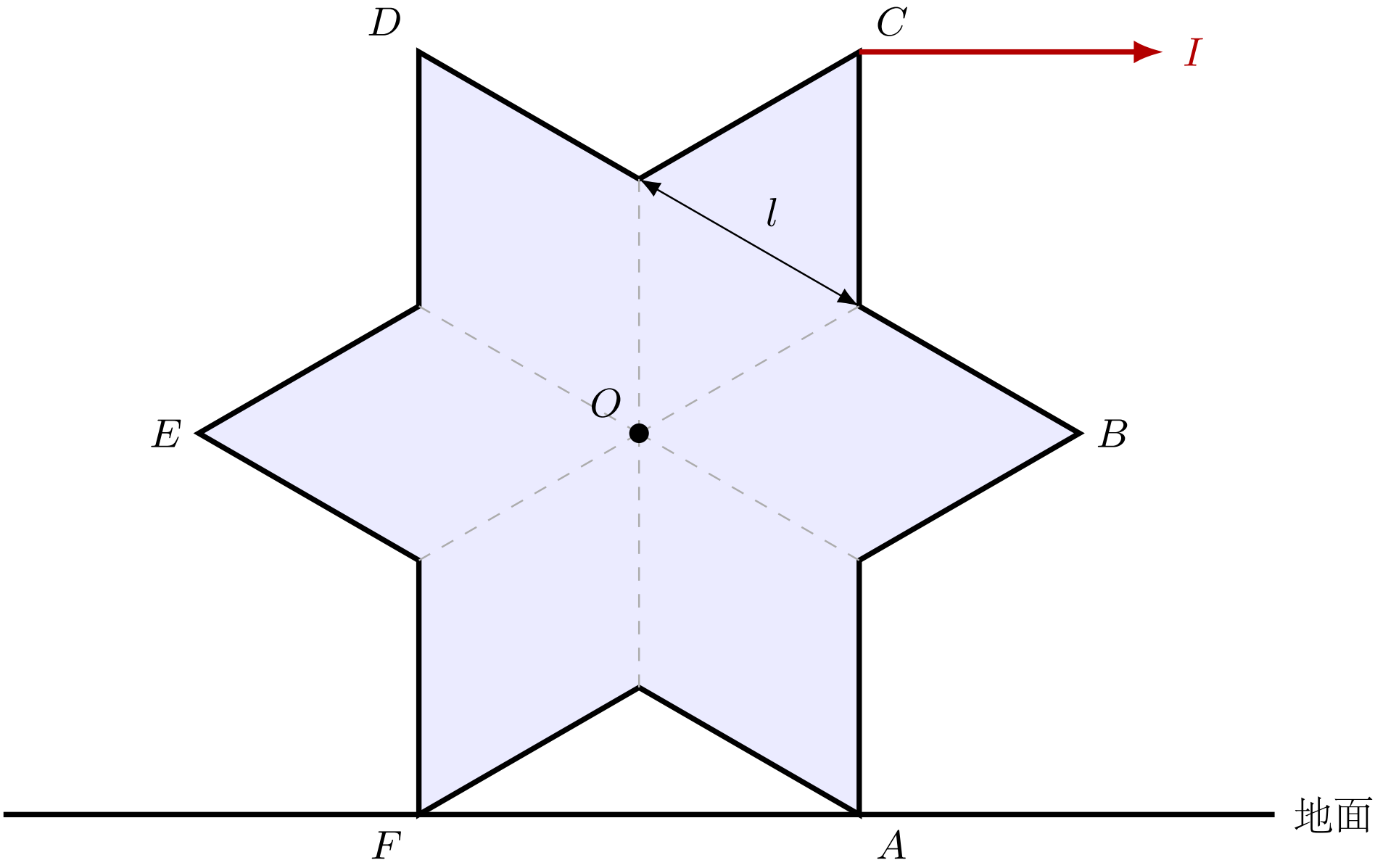
$\theta = \pi/2$

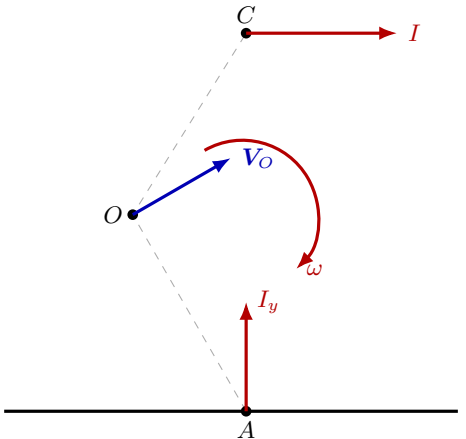


$\theta = \pi$

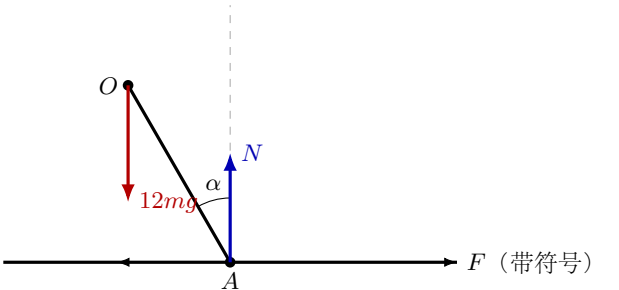


$\theta = 3\pi/2$

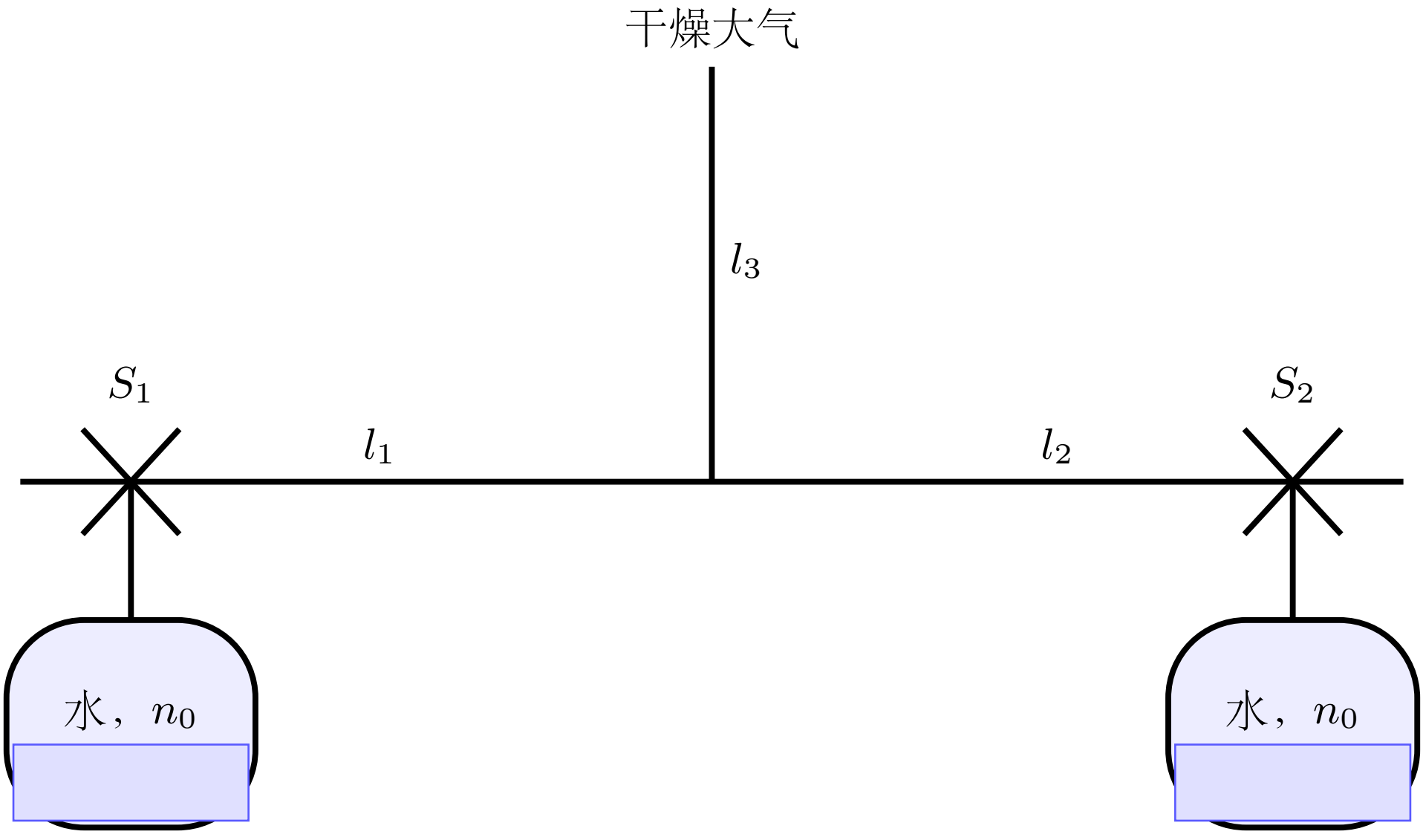

D
C
I
l
O
E
B
地面
F
A

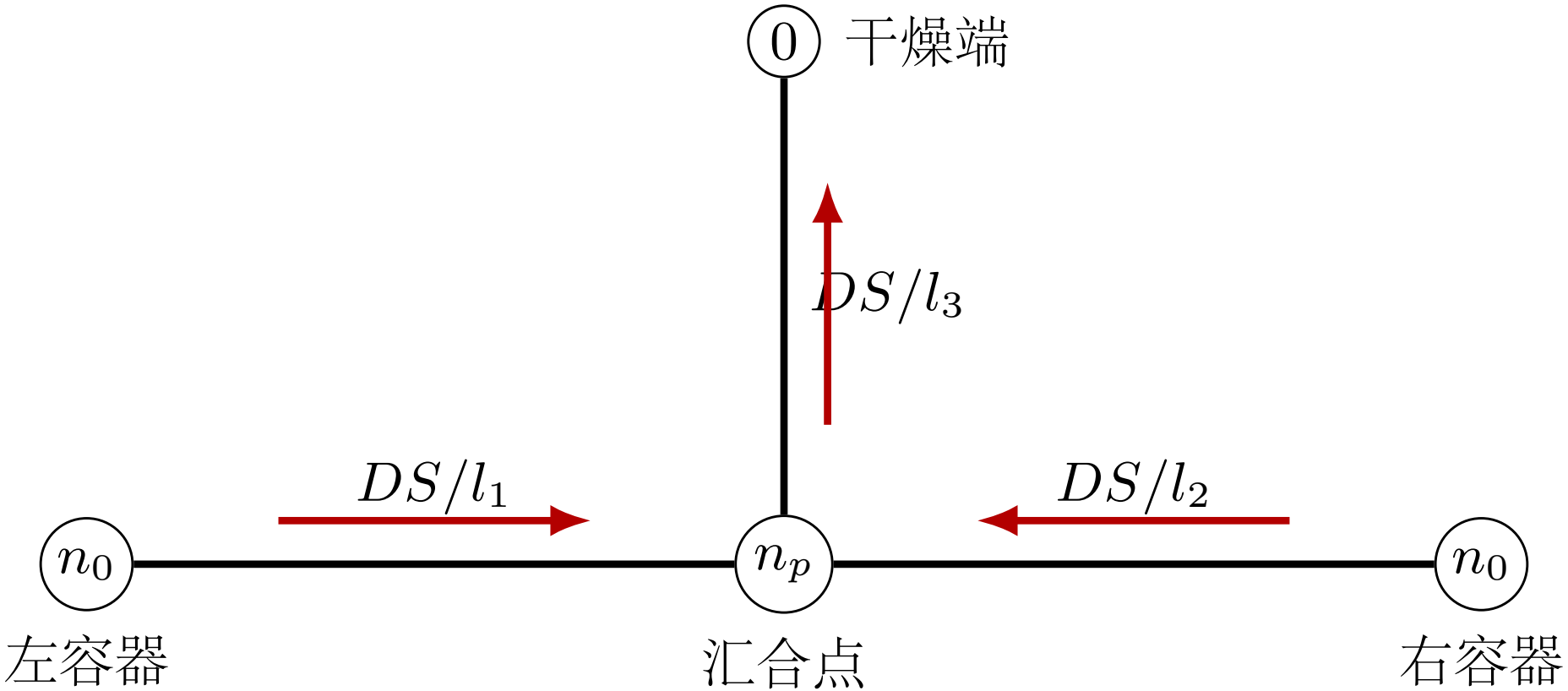


$OA = \sqrt{3}l, \quad AC = 3l$

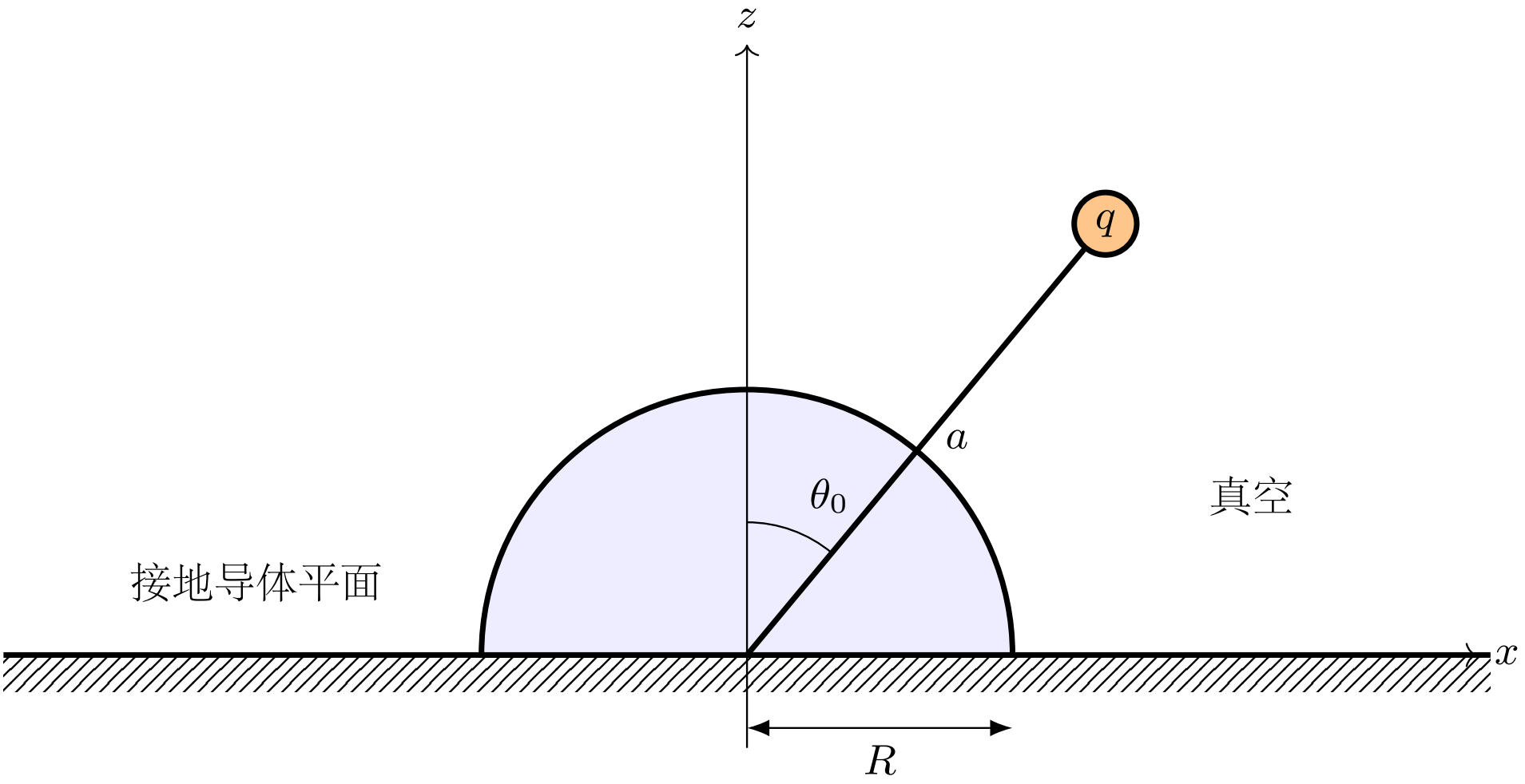


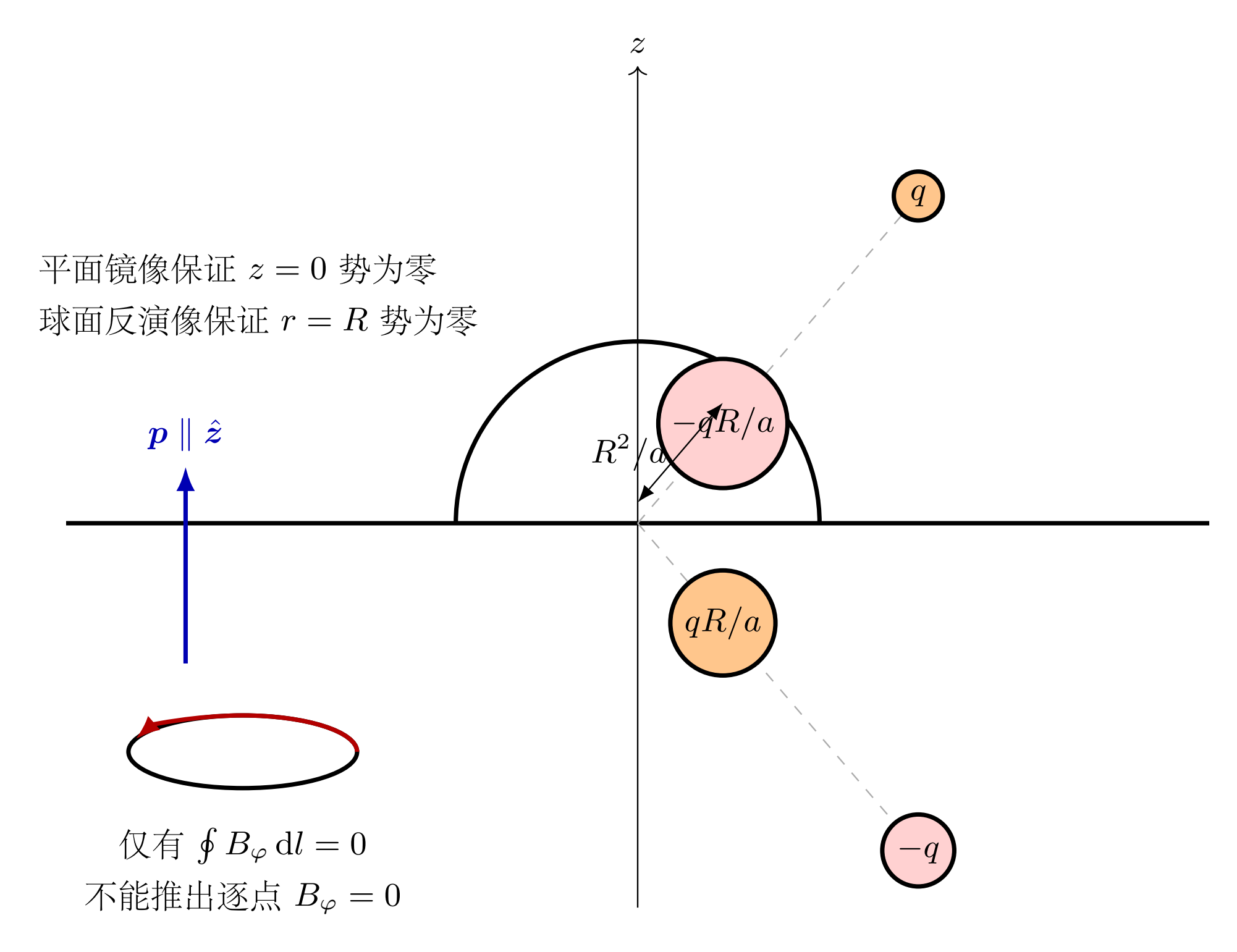
绕 A 翻越时检验能量、$N \geq 0$ 与 $|F| \leq \mu N$

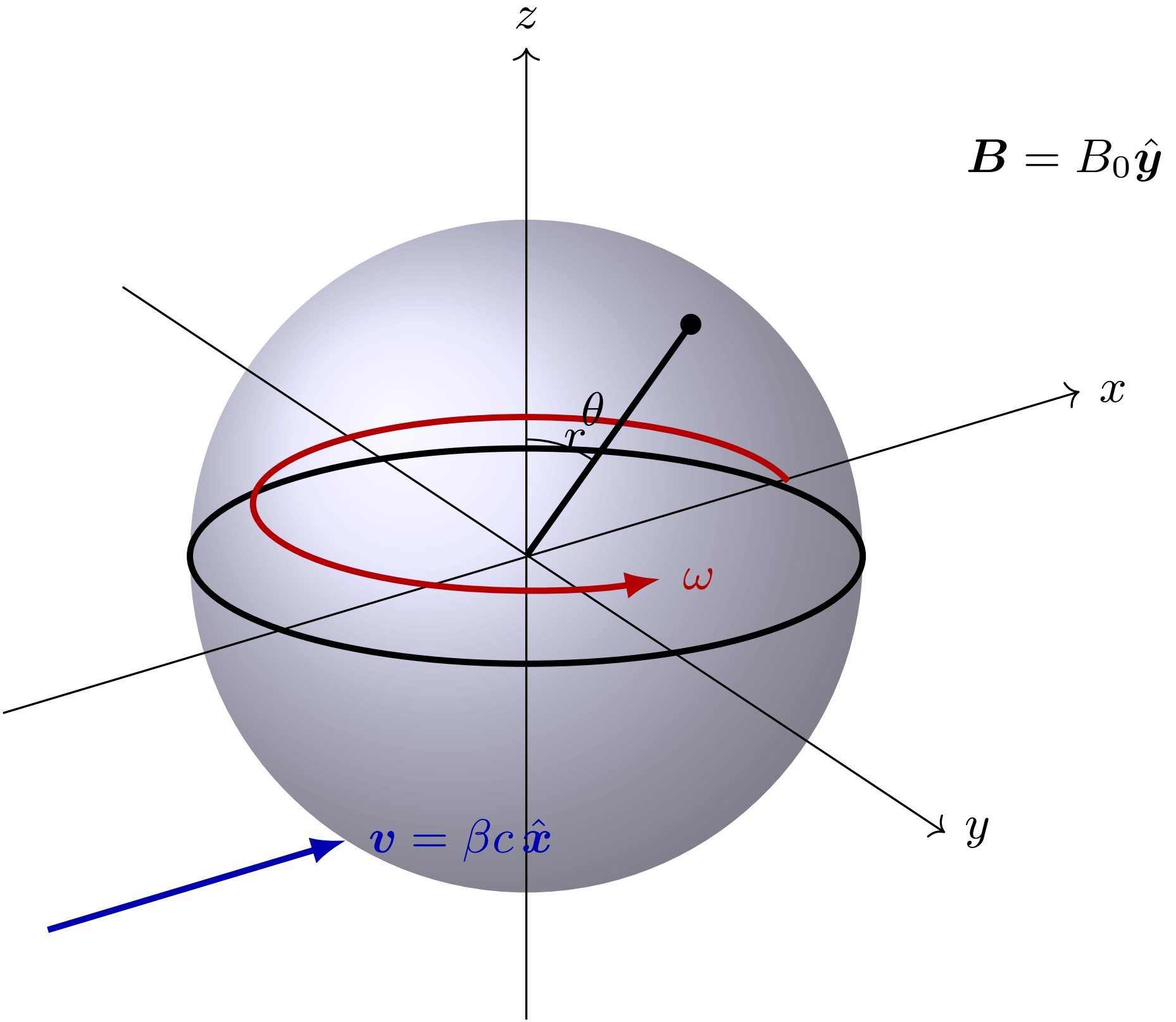

干燥大气
l_3
S_1
S_2
l_1
l_2
水，n_0
水，n_0



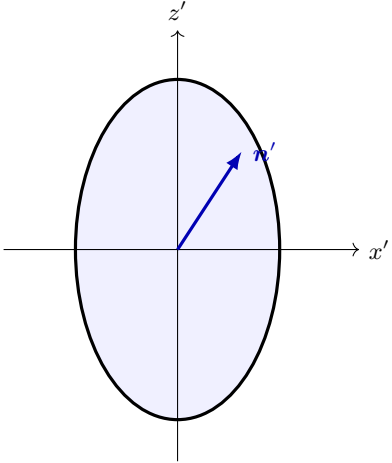
$$\frac{n_0 - n_p}{l_1} + \frac{n_0 - n_p}{l_2} = \frac{n_p}{l_3}$$

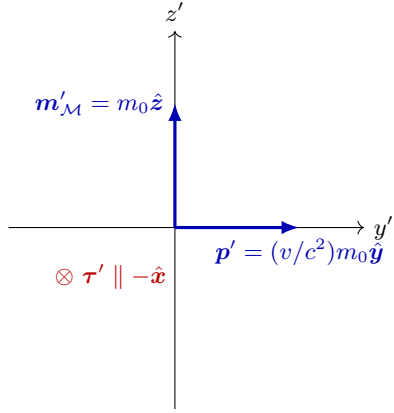

z
q
a
θ0
真空
接地导体平面
x
R


z
q
平面镜像保证 $z=0$ 势为零
球面反演像保证 $r=R$ 势为零
$-qR/a$
R^2/a
$\boldsymbol{p} \parallel \hat{\boldsymbol{z}}$
qR/a
$-q$
仅有 $\oint B_\varphi \,\mathrm{d}l = 0$
不能推出逐点 $B_\varphi = 0$


z
$\boldsymbol{B} = B_0\hat{\boldsymbol{y}}$
x
θ
r
ω
y
$\boldsymbol{v} = \beta c\,\hat{\boldsymbol{x}}$

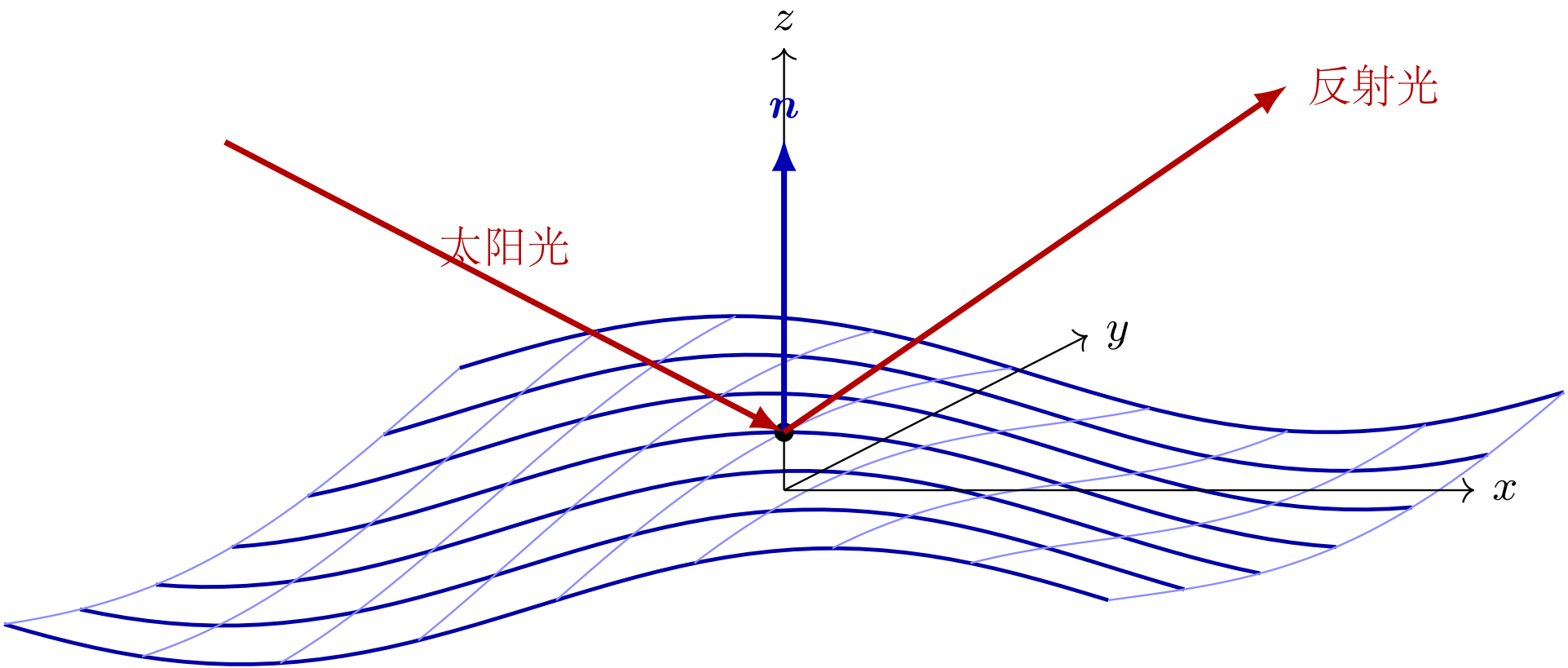
电流积分磁矩：$m'_j = m_0(1-\beta^2/2)$；须区分定义



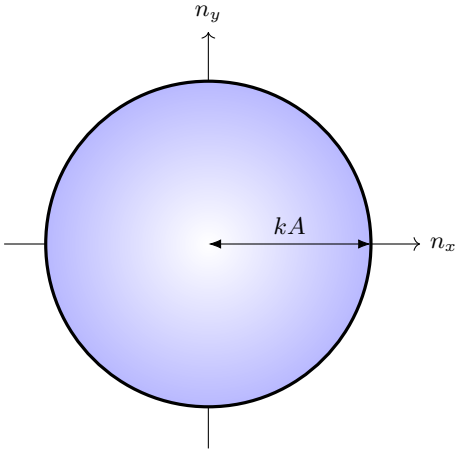


$x'^2/(r/\gamma)^2 + y'^2/r^2 + z'^2/r^2 = 1$

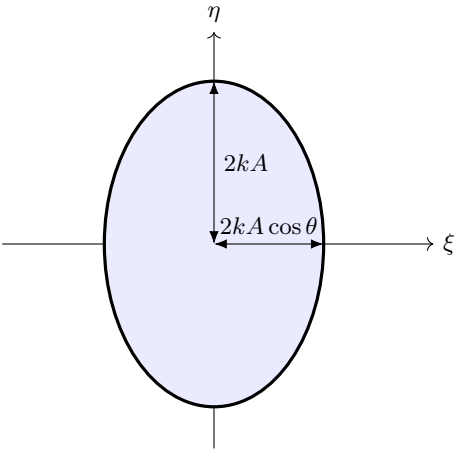
$\boldsymbol{B}' = \gamma B_0\hat{\boldsymbol{y}}, \quad \boldsymbol{E}' = \gamma v B_0\hat{\boldsymbol{z}}$



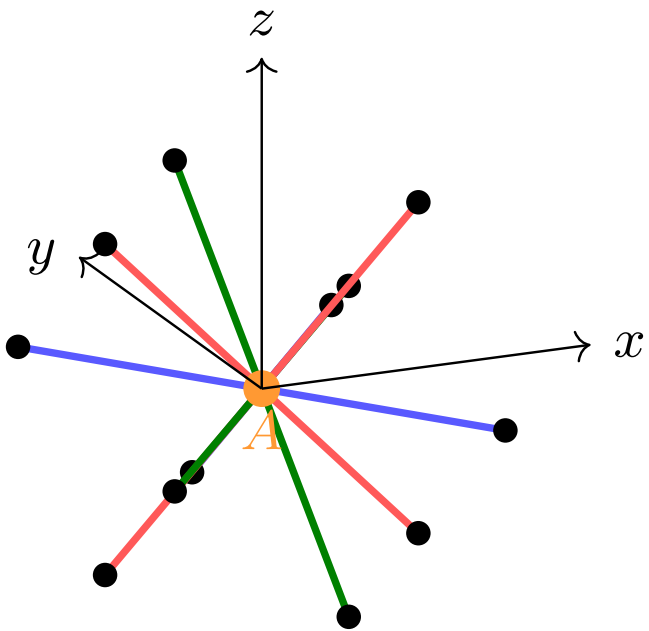
$$z(x,y) = A\cos(k_x x + k_y y - \omega t)$$



$f(n_x, n_y) \propto [\rho\sqrt{(kA)^2 - \rho^2}]^{-1}$



$\xi = 2n_x \cos\theta, \quad \eta = 2n_y$



十二个最近邻：

$(\pm l/2, \pm l/2, 0)$

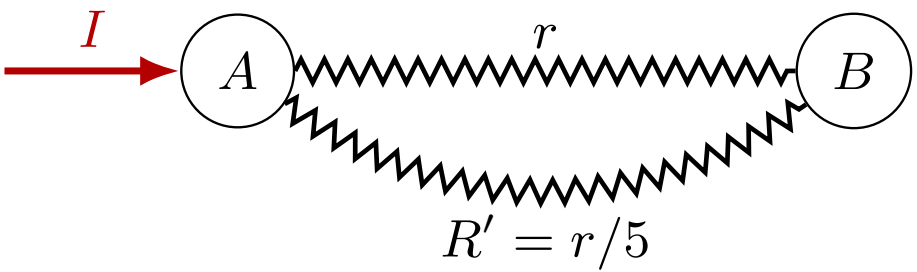
$(\pm l/2, 0, \pm l/2)$

$(0, \pm l/2, \pm l/2)$

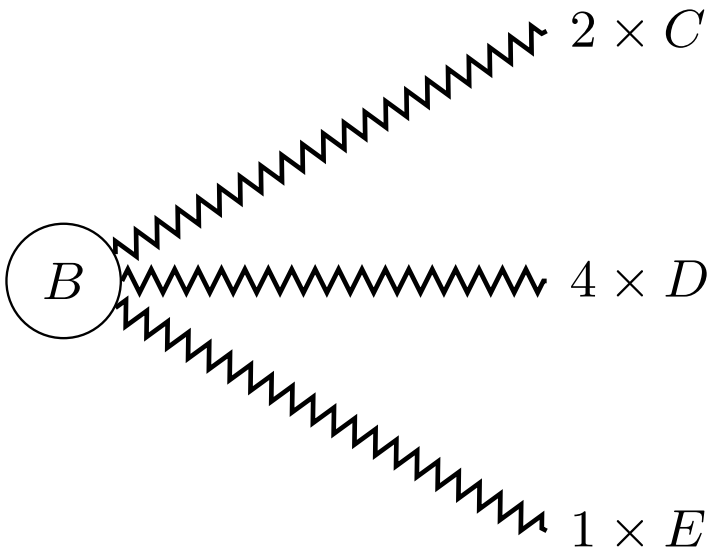
$B = (l/2, l/2, 0),\ C = (l, 0, 0)$

$D = (l, l/2, l/2),\ E = (l, l, 0)$

每条最近邻连线接电阻 r



$$R_{AB} = r \parallel (r/5) = r/6$$



其余四个邻点与 B 等势

$$I/12 = 2I_{BC} + 4I_{BD} + I_{BE}$$

$$2R_{AC} + 4R_{AD} + R_{AE} = 4r/3$$